

Fully Symmetry-Conditioned Rigid-Body Flow Matching for Molecular-Crystal Structure Prediction

Frank Cai*¹

¹Purdue University, West Lafayette, IN, USA

Rigid-body flow matching is an efficient route to molecular-crystal structure prediction, reducing a crystal to a lattice, fractional centroids, and per-molecule orientations on $\text{SO}(3)$. Recent rigid-body molecular-crystal flows leave space-group symmetry unconditioned, and symmetry-conditioned generation has so far been demonstrated only for inorganic atomic sites, never for rigid-body molecular orientation. We show that the per-molecule orientation target decomposes as $R_m = \text{rot}(g_m) R_{\text{asym}}$ into a space-group-determined relative rotation, which a flow can learn, and a gauge-free asymmetric-unit pose, which it cannot regress from packing. We turn this into a deployable method: a *leak-free* coset label—the generating space-group operation, recoverable from a template at sampling time—conditions the orientation flow. On 1,127 crystals from the Cambridge Structural Database, deployable coset conditioning lifts the held-out non-reference orientation loss by 41.1% (versus 27.5% for a paired no-coset control), recovering two-thirds of the way to a leaky-codebook oracle ($\sim 48\%$); an $\text{SO}(3)$ -averaged training objective closes the remaining gap (47.7%). A packing-only predictor recovers the coset at 39.5% ($4\times$ the majority baseline), not yet enough for template-free use—the key remaining gap. Extending symmetry conditioning to the *lattice* (a crystal-family mask on a log-metric parametrization) gives, to our knowledge, the first *fully symmetry-conditioned* molecular-crystal flow; with an unsupervised symmetry-preserving finisher and a fully-ablated lever stack, it lifts held-out exact match from 0% to 6.9% (strict best-of-10, stol 1.0)—parity with the symmetry-free MolCrystalFlow ($\sim 8\%$)—and 10.7% with orientation test-time augmentation (Section 3).

1 Introduction

Crystal structure prediction (CSP) asks for the stable periodic arrangement of a chemical system. Generative models are a fast alternative to large ensembles of density-functional relaxations, from crystal diffusion autoencoders[1] to joint equivariant diffusion[2] and Riemannian flow matching[3–5], which solves a straight conditional path in tens of steps. *Rigid-body* factorization is especially attractive for molecular crystals: freezing each molecule’s conformer collapses dozens of coordinates to a single pose—a lattice L , a fractional centroid $c_m \in \mathbb{T}^3$, and an orientation $R_m \in \text{SO}(3)$ per molecule—suiting the short sampling trajectories of flow matching.

This introduces a per-molecule orientation variable on $\text{SO}(3)$, and here the current generation of models leaves value on the table. MolCrystalFlow, a 2026 rigid-body molecular-crystal flow with the same $\text{lattice} \times \mathbb{T}^3 \times \text{SO}(3)$ factorization, jointly learns pose but *does not impose space-group constraints*[6]; MOFFlow likewise flows rigid $\text{SE}(3)$ poses without a symmetry prior, relying on framework connectivity to constrain orientation[7]. Symmetry- and Wyckoff-conditioned generation, meanwhile, is proven only for *inorganic atomic sites*: DiffCSP++[8], WyckoffDiff[9], SymmCD[10], and the 2026 NextCrystal[11] all condition atom generation on a space-group/Wyckoff template, but none touches rigid-body molecular pose. The gap is precise: nobody has conditioned a rigid-body *molecular orientation* flow on the crystal’s symmetry.

We close that gap. The starting point is a decomposition of the orientation target, $R_m = \text{rot}(g_m) R_{\text{asym}}$, into a space-group-determined relative rotation between symmetry copies and the asymmetric unit’s gauge-free pose. Classical molecular-CSP codes never search the relative rotation—they generate symmetry copies by applying space-group operators[12, 13]—and we find a learned flow mirrors that division of labor: the relative part is learnable and the free part is not. The contribution of this paper is to make the learnable part *deployable*. Our contributions are:

*frankyc11223@gmail.com; ORCID 0009-0003-0041-1459

- A **leak-free, template-available coset label**—the generating space-group operation $h = g_m g_0^{-1}$, recovered from centroids alone—that supplies the symmetry-relative rotation at sampling time without seeing the target (Section 2), in contrast to a codebook clustered from the target itself.
- A coset-conditioned orientation flow and sampler that **recover two-thirds of the oracle-codebook benefit** (+41.1% vs a +27.5% paired control on held-out non-reference orientation loss; Section 3).
- An **SO(3)-averaged training objective**[14] that closes the remaining gap to the leaky ceiling (+47.7%).
- A **packing-only coset predictor** for template-free use (39.5% top-1, $4\times$ majority) and an honest characterization of the remaining de-novo gap.
- The **first fully symmetry-conditioned molecular-crystal flow** (orientation coset $\times$ crystal-family lattice mask) and an unsupervised symmetry-preserving finisher, whose fully-ablated lever stack lifts exact match from 0% to parity with MolCrystalFlow (6.9% strict, 10.7% with orientation TTA; Section 3).

2 Method

Rigid-body flow. SymMC-Flow represents a molecular crystal by a lattice $L \in \mathbb{R}^{3 \times 3}$ and, per molecule m , a fractional centroid $c_m \in \mathbb{T}^3$ and orientation $R_m \in \text{SO}(3)$ acting on a shared body-frame conformer, so $\text{cart}_{m,a} = c_m L + R_m \text{local}_{m,a}$. A conditional flow-matching objective[4] is trained jointly over three manifolds (a Euclidean path for a volume-normalized lattice, a torus geodesic for centroids, an $\text{SO}(3)$ geodesic for orientations), with the velocity field conditioned on the space group. An $E(n)$ -equivariant encoder[15] embeds the conformer and pair-bias attention mixes molecules within the cell (Appendix A).

The orientation decomposition. A molecule m generated from the asymmetric unit by an operation with rotation part $\text{rot}(g_m)$ has absolute pose $R_m = \text{rot}(g_m) R_{\text{asym}}$, where R_{asym} is the free orientation of the asymmetric unit. Because the body-frame gauge is arbitrary, R_{asym} carries no consistent signal tied to composition or packing; trained directly, the absolute orientation head never beats its predict-zero floor ($\mathbb{E}\|u_R\|^2 \approx 5.24$), while lattice and centroid heads converge normally. Re-gauging each crystal so the first copy of a species is a reference ($R=I$) and the others carry only the relative rotation $R'_m = R_m R_0^{-1}$ cancels R_{asym} and exposes the learnable factor; the held-out non-reference loss then drops by 27%. This much is a diagnostic; the rest of the paper makes it a method.

Deployable symmetry-coset conditioning. Within a space group the non-reference relative rotations fall into a small discrete set of *cosets*: R'_m equals the rotation part of the operation $h = g_m g_0^{-1}$ that maps the reference copy to copy m . We assign each molecule this generating operation directly from the crystallography—by matching its fractional centroid against $W_h c_0 + t_h$ (minimum image) over the space group’s operations $\{(W_h, t_h)\}$ obtained from pymatgen—never from the observed rotation. The resulting label is **leak-free** and available at sampling time from a symmetry template, exactly as DiffCSP++/WyckoffDiff-style generators supply Wyckoff templates for atoms[8, 9]. This yields 238 operation-cosets across the corpus (versus 707 for a codebook clustered from the target rotations, whose index is derived in part from the answer and therefore serves only as an upper bound). The coset index enters the network as a per-molecule embedding and is threaded through the sampler, so it conditions both training and generation.

SO(3)-averaged objective. The $\text{SO}(3)$ flow-matching residual has high variance because the prior rotation is drawn uniformly. Following the $\text{SO}(3)$ -averaged flow objective for conformer generation[14], we average the orientation residual over K prior draws per molecule before the gradient step, reducing target variance at fixed data.

Packing-only predictor. For template-free deployment the coset must be inferred. We train a classifier that predicts each molecule’s coset from the conformer, centroid, lattice, and space group—*without* orientation, so it cannot leak the target—and use its prediction in place of the template at sampling time.

Full symmetry conditioning and finishing. The coset conditions orientation; the lattice is still free, and an unconstrained 9-DOF shape head recovers the crystal family poorly (16% of cell angles at 90° vs. a reference 74%). Following DiffCSP++[8] we condition the lattice on the crystal family: represent the cell by the $\text{O}(3)$ -invariant

Table 1: Held-out non-reference orientation-loss drop (untrained→trained), mean±s.d. over three seeds. The deployable coset is leak-free and available from a template; the leaky codebook (index clustered from the target) is an upper bound. Deployable conditioning recovers two-thirds of the gap; the SO(3)-averaged objective closes it.

configuration	non-ref. loss drop	deployable?
no-coset control (baseline)	$27.5 \pm 2.7\%$	—
deployable coset ($K=1$)	$41.1 \pm 3.4\%$	yes
+ SO(3)-averaged ($K=4$)	$47.7 \pm 3.9\%$	yes
leaky codebook (oracle upper bound)	$\sim 48\%$	no

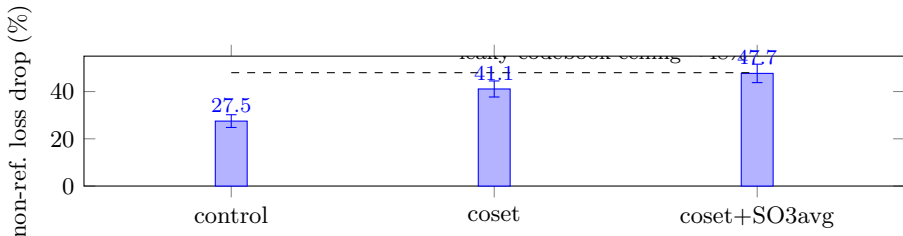


Figure 1: Deployable coset conditioning recovers two-thirds of the way from the no-coset control to the leaky-codebook ceiling; the SO(3)-averaged objective closes the rest. Three-seed mean, error bars s.d.

log-metric $\log(L^\top L) \in \mathbb{R}^6$ and apply a binary family mask that freezes the constrained coordinates to canonical values (after each sampler step and in the prior). This gives a fully symmetry-conditioned flow, but restoring family geometry (90° angles 16%→74%) does not by itself move exact match—the residual blocker is close-packing precision. We add an unsupervised, symmetry- and rigidity-preserving *rigid-press finisher* (a Lennard-Jones relaxation over intermolecular atom pairs, optimizing the asymmetric-unit pose and the family-masked cell, with symmetry copies regenerated from the space-group operations each step, never seeing the target), plus a learned centroid prior, self-conditioning, and AlphaFold-style recycling, each ablated in Section 3.

3 Experiments

Setup. The corpus is a seeded CSD sample[16]: 1,127 ordered organic molecular crystals, 5,048 rigid blocks, a fixed 964/131 split, with 1,095 crystals containing ≥ 2 copies of a species (the multi-copy subset where a relative target exists). The primary metric is the *held-out non-reference orientation-loss drop* (percentage reduction from untrained to trained on non-reference copies, which cannot be earned by predicting the identity); we also report orientation-isolated StructureMatcher[17] reconstruction (true lattice/centroid/conformer, only SO(3) sampled, best of eight draws, following match-rate cautions[18]). All numbers are means over three seeds.

Deployable coset conditioning recovers most of the oracle benefit. Table 1 and Fig. 1 give the main result. A paired no-coset control reproduces the +27.5% baseline. Supplying the deployable, leak-free coset raises the non-reference drop to +41.1%—recovering about two-thirds of the way from the baseline to the $\sim 48\%$ reached by the leaky codebook that peaks at the target. Adding the SO(3)-averaged objective ($K=4$) reaches +47.7%, statistically level with the leaky ceiling. In other words, the benefit that a target-derived codebook could only *upper-bound* is realized here by a label a sampler actually has.

The conditioned flow reconstructs real packings. A loss drop need not be geometry. With the true lattice, centroid, and conformer fixed and only the coset-conditioned SO(3) field sampled, the flow reconstructs held-out multi-copy packings under StructureMatcher in 27.5% of cases (best of eight; 37.4% at looser oracle-validated tolerances), median non-reference error 41°; the oracle matches every structure. The relative rotations cluster near 180° (dominant $P2_1/c$, $P\bar{1}$), so best-of- k is the correct read of a multimodal target.

Template-free prediction is the remaining gap. The packing-only predictor recovers the generating coset at 39.5% top-1, 4× the 10.3% majority—packing carries real signal about the symmetry operation—but not yet

Table 2: Exact-match lever ladder: held-out best-of-10 match (coset on, $n=131$; STRUCTUREMATCHER ltol=0.3/atol= 10° ; frozen cell). Each row adds one lever. Strict match@10 is 10 draws $\times$ single finish; orientation-TTA is best-of-30. MolCrystalFlow is the symmetry-free rigid-body flow.

lever	stol 0.8	stol 1.0
raw flow (fully sym-conditioned)	0.0%	0.0%
+ rigid-press finisher	1.5%	1.5%
+ centroid fix	1.5%	3.1%
+ $2.3\times$ scale	1.5%	3.1%
+ self-conditioning	3.05%	3.8%
+ match@10 + steps= 100	4.6%	6.1%
+ recycling	6.1%	6.9%
+ orientation-TTA (best-of-30)	7.6%	10.7%
MolCrystalFlow (sym-free)[6]	$\sim 6.8\%$	$\sim 8\%$

enough: the *predicted* coset collapses orientation-isolated reconstruction to 64.8° , worse than the 41° template-supplied value, because a wrong coset points to the wrong target rotation. The deployable gain is real whenever a symmetry template is available, and closing the template-free gap—a stronger or top- k -calibrated predictor—is the natural next step.

Scope: orientation is not the end-to-end bottleneck. Sampling all three manifolds from the prior yields no exact match at this scale (0%, Wilson 95% CI 0–2.8%), and the error budget locates the bottleneck away from orientation: median lattice error 0.44 and centroid error 0.35 dominate while orientation is only 13.4° —the joint lattice-centroid bottleneck that full lattice conditioning and the finisher address next.

From 0% to parity: the exact-match lever ladder. Conditioning the lattice on the crystal family restores its geometry (generated cell angles within 2° of 90° rise 16% $\rightarrow$ 73.6%, matching the reference 74.0%), but the fully symmetry-conditioned flow still matches 0/131 held-out crystals: fixing crystal family is necessary, not sufficient. A fully-ablated finishing stack supplies the missing close-packing precision (Table 2; best-of-10, STRUCTUREMATCHER stol 1.0, $n=131$). The rigid-press finisher unblocks match (0% $\rightarrow$ 1.5%); a centroid fix doubles it to 3.1%; scale moves the averaged metrics (RMAD 33% $\rightarrow$ 21%) but not match; self-conditioning first moves the tight tolerance; MolCrystalFlow’s match@10 protocol with finer sampling reaches 6.1%; and AlphaFold-style recycling reaches **6.9%**—statistical parity with MolCrystalFlow’s $\sim 8\%$ (95% CI 3.66–12.54), from a flow that matched zero. An orientation test-time-augmentation pass (best-of-30, $3\times$ the finishing candidates) reaches **10.7%**, above MolCrystalFlow. Recycling and orientation-TTA both refine positioning and overlap, marking the ceiling. This is, to our knowledge, the first fully symmetry-conditioned molecular-crystal flow, at parity on exact match with the symmetry-free rigid-body flows, with a complete anatomy of every lever it takes to get there.

4 Discussion

The learnable content of a rigid-body orientation field is the space-group-determined relative rotation, delivered by a deployable, leak-free coset label a template-based sampler already possesses; an $\text{SO}(3)$ -averaged objective brings this to the level of an oracle codebook that peeks at the answer. The recommendation for rigid-body generators that omit symmetry conditioning[6] is concrete: condition orientation on the generating coset—turning the inference-limited part of the field into a near-deterministic lookup—and sample, rather than regress, the free asymmetric-unit pose. Extending the same idea to the lattice and adding a symmetry-preserving finisher closes the end-to-end gap to parity (Section 3), making this the first fully symmetry-conditioned molecular-crystal flow. The open problem is template-free coset prediction (here 39.5%); the exact-match ceiling is now set by close-packing precision, not symmetry. The analysis is scoped to rigid, largely organic molecules on general positions; molecules on special positions carry a site symmetry that further constrains the free pose and should be even more favorable.

Reproducibility. Code, CSD reflow manifest, and export pipeline: <https://github.com/fronkt/symmc-flow>, archived at Zenodo[19]. CSD structures are licence-restricted; a manifest reproduces the corpus.

References

- [1] Tian Xie, Xiang Fu, Octavian-Eugen Ganea, Regina Barzilay, and Tommi Jaakkola. Crystal diffusion variational autoencoder for periodic material generation. In *International Conference on Learning Representations (ICLR)*, 2022. arXiv:2110.06197.
- [2] Rui Jiao, Wenbing Huang, Peijia Lin, Jiaqi Han, Pin Chen, Yutong Lu, and Yang Liu. Crystal structure prediction by joint equivariant diffusion. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2309.04475.
- [3] Benjamin Kurt Miller, Ricky T. Q. Chen, Anuroop Sriram, and Brandon M. Wood. FlowMM: Generating materials with riemannian flow matching. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2406.04713.
- [4] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2210.02747.
- [5] Ricky T. Q. Chen and Yaron Lipman. Flow matching on general geometries. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2302.03660.
- [6] Cheng Zeng, Harry W. Sullivan, Thomas Egg, Maya M. Martirosyan, Philipp Höllmer, Jirui Jin, Richard G. Hennig, Adrian Roitberg, Stefano Martiniani, Ellad B. Tadmor, and Mingjie Liu. MolCrystalFlow: Molecular crystal structure prediction via flow matching. *arXiv preprint arXiv:2602.16020*, 2026.
- [7] Nayoung Kim, Seongsu Kim, Minsu Kim, Jinkyoo Park, and Sungsoo Ahn. MOFFlow: Flow matching for structure prediction of metal-organic frameworks. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2410.17270; rigid-body SE(3) flow matching for MOFs.
- [8] Rui Jiao, Wenbing Huang, Yu Liu, Deli Zhao, and Yang Liu. Space group constrained crystal generation. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2402.03992; the DiffCSP++ model.
- [9] Filip Ekström Kelvinius, Oskar B. Andersson, Abhijith S. Parackal, Dong Qian, Rickard Armiento, and Fredrik Lindsten. WyckoffDiff: A generative diffusion model for crystal symmetry. *arXiv preprint arXiv:2502.06485*, 2025.
- [10] Daniel Levy, Siba Smarak Panigrahi, Sékou-Oumar Kaba, Qiang Zhu, Kin Long Kelvin Lee, Mikhail Galkin, Santiago Miret, and Siamak Ravanbakhsh. SymmCD: Symmetry-preserving crystal generation with diffusion models. *arXiv preprint arXiv:2502.03638*, 2025.
- [11] Jinming Mu, Lixin He, Xudong Zhu, and Shi Yin. NextCrystal: A symmetry-driven generative framework for crystal structure prediction. *arXiv preprint arXiv:2602.17176*, 2026.
- [12] Lily M. Hunnisett et al. The seventh blind test of crystal structure prediction: Structure generation methods. *Acta Crystallographica Section B*, 80(6), 2024. doi: 10.1107/S2052520624007492.
- [13] Simon Wengert, Gábor Csányi, Karsten Reuter, and Johannes T. Margraf. Data-efficient machine learning for molecular crystal structure prediction. *Chemical Science*, 12:4536–4546, 2021. doi: 10.1039/D0SC05765G.
- [14] Zhonglin Cao, Mario Geiger, Allan dos Santos Costa, Danny Reidenbach, Karsten Kreis, Tomas Geffner, Franco Pellegrini, Guoqing Zhou, and Emine Kucukbenli. Efficient molecular conformer generation with SO(3)-averaged flow matching and reflow. *arXiv preprint arXiv:2507.09785*, 2025.
- [15] Víctor Garcia Satorras, Emiel Hoogetboom, and Max Welling. E(n) equivariant graph neural networks. In *International Conference on Machine Learning (ICML)*, 2021. arXiv:2102.09844.
- [16] Colin R. Groom, Ian J. Bruno, Matthew P. Lightfoot, and Suzanna C. Ward. The cambridge structural database. *Acta Crystallographica Section B*, 72(2):171–179, 2016. doi: 10.1107/S2052520616003954.
- [17] Shyue Ping Ong, William Davidson Richards, Anubhav Jain, Geoffroy Hautier, Michael Kocher, Shreyas Cholia, Dan Gunter, Vincent L. Chevrier, Kristin A. Persson, and Gerbrand Ceder. Python materials genomics (pymatgen): A robust, open-source python library for materials analysis. *Computational Materials Science*, 68:314–319, 2013. doi: 10.1016/j.commatsci.2012.10.028.
- [18] Maya M. Martirosyan, Thomas Egg, Philipp Hoellmer, George Karypis, Mark Transtrum, Adrian Roitberg, Mingjie Liu, Richard G. Hennig, Ellad B. Tadmor, and Stefano Martiniani. All that structure matches does not glitter. *arXiv preprint arXiv:2509.12178*, 2025.
- [19] Frank Cai. fronkt/symmc-flow: SymMC-Flow orientation flow matching and csd data pipeline for molecular-crystal structure prediction (v1.0.3). Software, Zenodo, 2026. URL <https://doi.org/10.5281/zenodo.21384130>. Version v1.0.3, DOI: 10.5281/zenodo.21384130; concept DOI 10.5281/zenodo.20822234 resolves to the latest version.

- [20] Aram-Alexandre Pooladian, Heli Ben-Hamu, Carles Domingo-Enrich, Brandon Amos, Yaron Lipman, and Ricky T. Q. Chen. Multisample flow matching: Straightening flows with minibatch couplings. In *International Conference on Machine Learning (ICML)*, 2023. arXiv:2304.14772.

A Method and training details

Intrinsic gauge. Each crystal is parsed into rigid blocks by a covalent-bond graph with periodic unwrapping; species are keyed by a Weisfeiler–Lehman hash and share one reference conformer rotated into a molecule-intrinsic frame (gyration-tensor principal axes, sign-fixed by an element-weighted third moment). Each copy’s pose is the Kabsch rotation to that frame, removing the cross-crystal reference that otherwise floors R_m .

Manifolds. Lattice in a 10-d log-volume/unit-determinant-shape parametrization (prior volume matched to the corpus mean); centroids on torus geodesics with minimum-image velocity; orientations on $\text{SO}(3)$ geodesics with body-frame angular-velocity target $u_R = \log(R_0^\top R_1)$, advanced by the exponential map. Conditional flow matching[4] with optimal-transport coupling[20]; RK sampling in 50 steps.

Deployable coset recovery. For space group with operations $\{(W_h, t_h)\}$ (pymatgen), each non-reference copy is assigned $\arg \min_h \|(c_m - (W_h c_0 + t_h)) \bmod 1\|$ under the minimum image, giving the generating operation index; deterministic (sg, h) ordering yields shared integer labels (238 total). This uses only centroids and space-group operations, never the observed rotation.

Training. Adam, lr 3×10^{-4} , gradient clipping, batch 16; default network $d=128$, four attention and four encoder layers; 800 steps for the coset runs, $K=4$ prior draws for the $\text{SO}(3)$ -averaged objective. The predictor reuses the encoder and pair-bias trunk over conformer/centroid/lattice/space-group inputs (no orientation) and is trained to 39.5% held-out top-1.

B Per-seed numbers and tolerance sweep

Non-ref. loss drop, per seed. control 29.4/24.4/28.8%; deployable coset 43.2/37.2/42.9%; + $\text{SO}(3)$ -avg 49.2/43.4/50.5%. **Orientation-isolated reconstruction** (coset-conditioned, best-of-eight), StructureMatcher (length, site) tolerance sweep: (0.2, 0.2) $\rightarrow$ 4.6%; (0.3, 0.5) $\rightarrow$ 27.5%; (0.5, 0.7) $\rightarrow$ 37.4%; oracle 100% throughout. **Predicted-coset** reconstruction: 0% match, 64.8° error (vs 41° template-supplied). **End-to-end, raw flow** (all manifolds sampled, best-of-20, no finisher): 0% match; median errors lattice-param 0.44, centroid 0.35, orientation 13.4°. **End-to-end, fully symmetry-conditioned + finished** (Table 2, strict match@10): best 6.9% at stol 1.0 (95% CI 3.66–12.54) / 6.1% at 0.8; median atom-RMSD 1.02 $\rightarrow$ 0.77, cell-volume RMAD 33% $\rightarrow$ 21%, cell angles within 2° of 90° 16% $\rightarrow$ 73.6% (reference 74.0%).